

Practical No: 12

Write a Program to create a text summarization tool.

```
In [2]: !pip install transformers  
        !pip install torch
```

Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages (4.51.3)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from transformers) (3.18.0)

Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.30.2)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (2.0.2)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (24.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from transformers) (6.0.2)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (2024.11.6)

Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from transformers) (2.32.3)

Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.21.1)

Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.5.3)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packages (from transformers) (4.67.1)

Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (2025.3.2)

Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (4.13.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (3.4.1)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (2025.1.31)

Requirement already satisfied: torch in /usr/local/lib/python3.11/dist-packages (2.6.0+cu124)

Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch) (3.18.0)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch) (4.13.2)

Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch) (3.4.2)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch) (3.1.6)

Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch) (2025.3.2)

Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch)

 Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)

Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch)

 Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)

Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch)

 Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)

Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch)

 Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)

Collecting nvidia-cublas-cu12==12.4.5.8 (from torch)

 Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata

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Collecting nvidia-cufft-cu12==11.2.1.3 (from torch)
 Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch)
 Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch)
 Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cuspars-cu12==12.3.1.170 (from torch)
 Downloading nvidia_cuspars-cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torch) (0.6.2)
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch) (2.21.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch) (12.4.127)
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch)
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Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch) (3.0.2)
Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl (363.4 MB)
  363.4/363.4 MB 4.4 MB/s eta 0:00:00
Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
  13.8/13.8 MB 85.8 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (24.6 MB)
  24.6/24.6 MB 66.8 MB/s eta 0:00:00
Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (883 kB)
  883.7/883.7 kB 53.1 MB/s eta 0:00:00
Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl (664.8 MB)
  664.8/664.8 MB 1.4 MB/s eta 0:00:00
Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl (211.5 MB)
  211.5/211.5 MB 7.6 MB/s eta 0:00:00
Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl (56.3 MB)
  56.3/56.3 MB 14.3 MB/s eta 0:00:00
Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl (127.9 MB)
  127.9/127.9 MB 7.3 MB/s eta 0:00:00
Downloading nvidia_cuspars-cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl (207.5 MB)
  207.5/207.5 MB 6.4 MB/s eta 0:00:00
Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
  21.1/21.1 MB 73.3 MB/s eta 0:00:00

Installing collected packages: nvidia-nvjitlink-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, nvidia-cusparse-cu12, nvidia-cudnn-cu12, nvidia-cusolver-cu12

Attempting uninstall: nvidia-nvjitlink-cu12

Found existing installation: nvidia-nvjitlink-cu12 12.5.82

Uninstalling nvidia-nvjitlink-cu12-12.5.82:

Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82

Attempting uninstall: nvidia-curand-cu12

Found existing installation: nvidia-curand-cu12 10.3.6.82

Uninstalling nvidia-curand-cu12-10.3.6.82:

Successfully uninstalled nvidia-curand-cu12-10.3.6.82

Attempting uninstall: nvidia-cufft-cu12

Found existing installation: nvidia-cufft-cu12 11.2.3.61

Uninstalling nvidia-cufft-cu12-11.2.3.61:

Successfully uninstalled nvidia-cufft-cu12-11.2.3.61

Attempting uninstall: nvidia-cuda-runtime-cu12

Found existing installation: nvidia-cuda-runtime-cu12 12.5.82

Uninstalling nvidia-cuda-runtime-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82

Attempting uninstall: nvidia-cuda-nvrtc-cu12

Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82

Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82

Attempting uninstall: nvidia-cuda-cupti-cu12

Found existing installation: nvidia-cuda-cupti-cu12 12.5.82

Uninstalling nvidia-cuda-cupti-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82

Attempting uninstall: nvidia-cublas-cu12

Found existing installation: nvidia-cublas-cu12 12.5.3.2

Uninstalling nvidia-cublas-cu12-12.5.3.2:

Successfully uninstalled nvidia-cublas-cu12-12.5.3.2

Attempting uninstall: nvidia-cusparse-cu12

Found existing installation: nvidia-cusparse-cu12 12.5.1.3

Uninstalling nvidia-cusparse-cu12-12.5.1.3:

Successfully uninstalled nvidia-cusparse-cu12-12.5.1.3

Attempting uninstall: nvidia-cudnn-cu12

Found existing installation: nvidia-cudnn-cu12 9.3.0.75

Uninstalling nvidia-cudnn-cu12-9.3.0.75:

Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75

Attempting uninstall: nvidia-cusolver-cu12

Found existing installation: nvidia-cusolver-cu12 11.6.3.83

Uninstalling nvidia-cusolver-cu12-11.6.3.83:

Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83

Successfully installed nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.2.1.3 nvidia-curand-cu12-10.3.5.147 nvidia-cusolver-cu12-11.6.1.9 nvidia-cusparse-cu12-12.3.1.170 nvidia-nvjitlink-cu12-12.4.127

```
In [2]: from transformers import pipeline

def summarize_text(text, max_length=150, min_length=50):
    summarizer = pipeline("summarization", model="facebook/bart-large-cnn")
    summary = summarizer(text, max_length=max_length, min_length=min_length, do_
    return summary[0]['summary_text']

# Example usage
long_text = """
Climate change is one of the most pressing issues facing our planet today.
```

It refers to long-term shifts in temperatures and weather patterns, mainly caused by human activities. These activities release greenhouse gases into the atmosphere, trapping heat and causing the Earth's average temperature to rise. The consequences of climate change are far-reaching and include more frequent and severe weather events, rising sea levels, and threats to biodiversity and food security.

```
summary = summarize_text(long_text)
print("Abstractive text summarization")
print("Original text length:", len(long_text))
print("Summary length:", len(summary))
print("\nSummary:")
print(summary)
```

/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:

The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart your session.

You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(

Device set to use cpu

Abstractive text summarization

Original text length: 843

Summary length: 307

Summary:

Climate change is one of the most pressing issues facing our planet today. It refers to long-term shifts in temperatures and weather patterns, mainly caused by human activities. These activities release greenhouse gases into the atmosphere, trapping heat and causing the Earth's average temperature to rise.

Device set to use cpu

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Climate change is one of the most pressing issues facing our planet today. It refers to long-term shifts in temperatures and weather patterns, mainly caused by human activities. These activities release greenhouse gases into the atmosphere, trapping heat and causing the Earth's average temperature to rise.